Problem 10.20

Fund A accumulates at a simple interest rate of 10%. Fund B accumulates at a simple discount rate of 5%. Find the point in time at which the force of interest on the two funds is equal.

Solution

For Fund A, the accumulation function under simple interest is

$$a_A(t) = 1 + 0.10t.$$

The force of interest is

$$\delta_A(t) = \frac{a'_A(t)}{a_A(t)} = \frac{0.10}{1 + 0.10t}.$$

For Fund B, the accumulation function under simple discount is

$$a_B(t) = \frac{1}{1 - 0.05t}, \qquad 0 \le t < 20.$$

Therefore,

$$\delta_B(t) = \frac{a'_B(t)}{a_B(t)} = \frac{0.05}{1 - 0.05t}.$$

Equating the two forces of interest gives

$$\frac{0.10}{1 + 0.10t} = \frac{0.05}{1 - 0.05t}.$$

Cross-multiplying,

$$\begin{aligned} 0.10(1 - 0.05t) &= 0.05(1 + 0.10t), \\ 0.10 - 0.005t &= 0.05 + 0.005t, \\ 0.05 &= 0.01t. \end{aligned}$$

Hence,

$$\boxed{t = 5 \text{ years}}.$$

At that time, the common force of interest is

$$\delta_A(5) = \delta_B(5) = \frac{0.10}{1.50} = \frac{1}{15} \approx 0.06667.$$